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# Lecture 1: Introduction to Psychology (PSY 101)

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## Class Norms and Expectations

### 1. Attendance & Participation

- Being present in class regularly is important because psychology often involves discussions, reflections, and participation-based learning activities.

### 2. Open-mindedness and Respect

- Since psychology explores human thoughts, emotions, and behavior, everyone may have different perspectives.

You must respect differing views and avoid judgment.

- ### 3. Encouragement to Question and Reflect
- Psychology thrives on curiosity. Asking “why” behind behaviors and mental processes helps deepen understanding.

### 4. Confidentiality

- During class discussions or personal sharing, maintaining privacy is essential to build trust and respect among peers.
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## Self-Reflection Before Beginning

Ask yourself:

- Why am I taking this course? (Interest, career, self-understanding?)
- What do I hope to learn? (About the brain, behavior, emotions?)
- How do I learn best? (Visually, auditorily, in groups, or independently?)

Self-awareness of your learning style can help you succeed in this course — just as psychology emphasizes self-understanding in general.

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## Course Overview

Psychology helps us understand **how and why humans think, feel, and behave** the way they do.

This course introduces both **scientific methods** and **theoretical foundations** of the field.

### Major Topics:

1. **Introduction to Psychology** – What it is and why it matters.
  2. **The Brain** – Structure and biological functions affecting behavior.
  3. **Perception** – How we interpret sensory information.
  4. **Cognition** – Thinking, reasoning, memory, and language.
  5. **Emotion** – Feelings and their impact on actions.
  6. **Personality** – Individual differences and traits.
  7. **Development** – Growth from infancy to adulthood.
  8. **Psychopathology** – Psychological disorders and mental illness.
  9. **Social & Group Interactions** – How we behave with others in groups and society.
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## Definition of Psychology

“Psychology is the **scientific study of behavior and mental processes.**”

Let's break this down:

| Term                    | Meaning                                                                                                                                                   |
|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Scientific</b>       | Psychology uses <b>systematic, objective, and empirical methods</b> (e.g., experiments, observations, surveys) to gather verifiable data.                 |
| <b>Behavior</b>         | Refers to <b>observable actions</b> like talking, walking, crying, eating — anything we can see or measure.                                               |
| <b>Mental Processes</b> | Refers to <b>internal experiences</b> such as thoughts, emotions, memory, and motives that cannot be directly observed but can be inferred from behavior. |

**Example:**

If someone suddenly cries, psychology tries to understand both the **behavior** (crying) and the **mental process** (sadness, loss, or stress) behind it.

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## History of Psychology

### Ancient Origins

Humans have always wondered about the mind and soul.

- In **Ancient Egypt**, the *Edwin Smith Papyrus* (around 1600 BCE) contained one of the earliest recorded observations about the **brain's role** in human functioning.
- Philosophers such as **Plato** and **Aristotle** in ancient Greece also debated the connection between the mind and body.

### Birth of Experimental Psychology

- **1854 – Gustav Fechner (Germany):**

Developed the **first theory** on how humans make judgments about sensory experiences.

→ His work laid the foundation for **Signal Detection Theory**, which explains how we detect faint signals amid background noise.

- **1879 – Wilhelm Wundt (Germany):**

Founded the **first psychology laboratory** in Leipzig.

→ Marked the official beginning of psychology as a **scientific discipline**.

→ Wundt focused on **introspection**, where people described their conscious experiences in detail.

→ He was the **first person to identify as a psychologist**.

## **Development and Global Spread**

- **G. Stanley Hall (1880s):**

- Established the first psychology lab in the **United States**.
- Promoted **scientific pedagogy** — applying psychology to education.

- **John Dewey (1890s):**

- Developed **educational theories** that emphasized **learning by doing** and adapting lessons to real-life experiences.
- Considered one of the fathers of modern education.

## **Behaviorism (Early 1900s)**

- Founded by **John B. Watson**, later expanded by **B. F. Skinner**.
- **Main Focus:** Study only **observable and measurable behavior** — not thoughts or feelings.
- Introduced the idea of **conditioning**:
  - **Classical Conditioning:** Learning through association (Pavlov's dogs).
  - **Operant Conditioning:** Learning through **rewards and punishments** (Skinner).

**Example:**

A student studies hard to get praise (reward) → increases studying behavior.

**Cognitive Revolution (Mid to Late 20th Century)**

- Psychologists began to realize that **mental processes** (thinking, memory, problem-solving) were crucial.
- This led to the rise of **Cognitive Science**, combining multiple disciplines:
  - Cognitive Psychology
  - Linguistics
  - Computer Science
  - Philosophy
  - Neurobiology

**Focus:** Understanding how we process information, think logically, store memories, and make decisions.

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**Major Perspectives in Psychology**

Each perspective provides a different way to explain human behavior:

| Perspective       | Focus Area                   | Key Assumption / Example                                                                                                                               |
|-------------------|------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Biological</b> | Brain, hormones,             | Behavior is influenced by biological structures and chemical processes. <i>Example:</i> Depression may be and genetics caused by low serotonin levels. |
| <b>Behavioral</b> | Learning from reinforcement. | Behavior is shaped through conditioning and environment. <i>Example:</i> A child learns to say “thank you” after being praised.                        |
| Perspective       | Focus Area                   | Key Assumption / Example                                                                                                                               |

|                       |                         |                                                                                                                                                             |
|-----------------------|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Cognitive</b>      | Mental processes        | Focuses on how people think, understand, and remember. <i>Example:</i> Problem-solving or decisionmaking strategies.                                        |
| <b>Psychoanalytic</b> | Unconscious motives     | Founded by Freud; behavior is driven by unconscious desires and childhood conflicts. <i>Example:</i> Phobias may reflect repressed fears.                   |
| <b>Humanistic</b>     | Free will and growth    | Emphasizes human potential, self-actualization, and personal growth. <i>Example:</i> People strive to become the best version of themselves.                |
| <b>Socio-cultural</b> | Society and culture     | Behavior is influenced by social norms, traditions, and cultural expectations. <i>Example:</i> Collectivist vs. individualist cultures affect family roles. |
| <b>Evolutionary</b>   | Adaptation and survival | Human traits evolved to enhance survival and reproduction. <i>Example:</i> Fear of snakes may be an evolved protective response.                            |

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## Perception, Thinking, and the Stroop Effect

### Automaticity

- When tasks become so familiar that they require **little conscious effort**.

*Example:* Reading is automatic — once you learn it, you don't need to think about every letter.\*

### Stroop Effect

- Demonstrates **interference between automatic and controlled processes**.
- Task: Name the **color** of the ink in which a word is printed rather than reading the word itself.

*Example:* The word "RED" written in blue ink.\*

- People take longer or make mistakes because **reading (automatic)** interferes with **color recognition (controlled)**.
  - This experiment shows that **attention** and **processing speed** affect performance.
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## Mental Health (According to WHO, 2022)

“Mental health is a state of mental well-being that enables people to cope with life’s stresses, realize their abilities, learn well and work well, and contribute to their community.”

### Key Takeaways:

1. Mental health ≠ just the absence of mental illness.  
→ It also means **positive psychological functioning**.
  2. Includes **emotional, psychological, and social well-being**.  
→ How we feel, think, and relate to others.
  3. Influenced by **individual, social, and structural factors** — such as genetics, environment, and community support.
  4. Mental health can **change over time**, depending on life circumstances.  
→ For instance, stress, loss, or trauma can affect well-being.
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## Objectives of Psychology

Psychology aims to **understand and explain human thought and behavior** through the following:

- **Describe** how people behave.
- **Explain** why they behave that way.

- **Predict** future behavior based on observation.
- **Control or influence** behavior positively (e.g., therapy, education, workplace training).

It studies:

- How our **minds interpret sensory information** (what we see, hear, taste, feel).
- How we **think, learn, and remember**.
- How we **interact socially** and **make moral or emotional decisions**.

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### Subfields of Psychology

| Subfield                                    | Focus Area                                                       | Example                                         |
|---------------------------------------------|------------------------------------------------------------------|-------------------------------------------------|
| <b>Biological Psychology</b>                | Links between brain, nervous system, and behavior                | Studying how brain injuries affect memory.      |
| <b>Clinical Psychology</b>                  | Diagnosis and treatment of disorders                             | Therapy for anxiety or mental depression.       |
| <b>Cognitive Psychology</b>                 | Study of mental processes like memory perception, memory, works. | Research on how reasoning                       |
| <b>Developmental Psychology</b>             | Growth and change across lifespan                                | Studying child development or aging.            |
| <b>Educational Psychology</b>               | Learning and teaching classroom strategies.                      | Designing effective methods                     |
| <b>Social Psychology</b>                    | Behavior in groups and social influence                          | Studying conformity or prejudice.               |
| <b>Industrial-Organizational Psychology</b> | Behavior in workplace settings                                   | Improving employee motivation and productivity. |
|                                             | Connection between mind                                          | Research on stress and its                      |



## Health Psychology

effects on the immune  
and physical health system.

Helping individuals achieve      Providing guidance for  
**Counseling Psychology** personal well-being      relationship or career issues.

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## In Summary

- **Psychology** = Science of **behavior and mental processes**.
  - It began as **philosophy**, became a **science** with Wundt in 1879, and evolved through **behaviorism** and **cognitive science**.
  - It includes multiple **perspectives** (biological, behavioral, cognitive, etc.), giving different explanations for human behavior.
  - It studies not only **mental illness** but also **positive mental health**, aiming to improve life quality.
  - Psychology applies to **education, health, relationships, workplaces, and society** as a whole.
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# Lecture 02: Research Methods in Psychology

## “Asking Questions, Getting Answers”

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### 1. What is Research in Psychology?

#### Definition

**Psychological research** is the **systematic, scientific study of behavior and mental processes**.

It uses **empirical methods** (observation, measurement, and experimentation) to understand how people think, feel, and act.

Psychological research aims to **develop theories, test hypotheses, and apply findings** to real-life problems such as mental health, education, workplace behavior, and social issues.

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#### Objectives of Psychological Research

| Goal           | Description                                                                 | Example                                                                |
|----------------|-----------------------------------------------------------------------------|------------------------------------------------------------------------|
|                | To observe and describe <b>Description</b> behavior as it occurs naturally. | Observing how children interact in a playground.                       |
|                | To determine <b>why</b> behavior <b>Explanation</b> occurs.                 | Studying whether aggression in children is caused by frustration.      |
|                | To predict how people will <b>Prediction</b> behave in certain situations.  | Predicting that sleep deprivation leads to lower academic performance. |
| <b>Control</b> | To influence or modify behavior. Using therapy to reduce anxiety.           |                                                                        |

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## Importance of Research in Psychology

1. **Evidence-based Understanding:** Replaces myths and opinions with tested knowledge.
2. **Improves Human Welfare:** Research findings improve education, health, business, and social policy.
3. **Theory Development:** Theories are built, tested, refined, or rejected through research.
4. **Scientific Progress:** Leads to innovation in therapy, learning methods, and diagnostics.
5. **Informs Practice:** Guides professionals like counselors, teachers, and clinicians.

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### Example:

Early research on **classical conditioning (Pavlov)** and **operant conditioning (Skinner)** laid the foundation for behavioral therapies still used today.

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## 2. What is Critical Thinking?

### Definition:

**Critical thinking** is the **intentional process of analyzing and evaluating information objectively** in order to make reasoned and evidence-based judgments.

It means not accepting statements at face value — instead, questioning assumptions, testing evidence, and considering alternative explanations.

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### Stephen D. Brookfield's (1987) Components of Critical Thinking:

1.     **Identifying and Challenging Assumptions** o     Recognize hidden biases, values, and beliefs behind arguments.
    - o     Example: “Men are naturally better at math” — a critical thinker questions whether this stereotype has scientific support.
  2.     **Recognizing the Importance of Context** o     Behavior and beliefs vary across situations and cultures. o     Example: Eye contact may show respect in one culture but rudeness in another.
  3.     **Imagining and Exploring Alternatives** o     Considering multiple explanations or solutions.
    - o     Example: A student scoring poorly might be unmotivated — or anxious, tired, or facing personal challenges.
  4.     **Reflective Skepticism** o     Withholding belief until sufficient evidence is available. o     Example: Questioning viral claims like “Listening to Mozart makes babies smarter.”
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### **Why Critical Thinking is Essential in Psychology**

- Prevents blind acceptance of “common sense” or “folk psychology.”
  - Encourages evidence-based reasoning.
  - Helps evaluate research validity and reliability.
  - Improves decision-making and ethical reasoning.
  - Encourages **scientific skepticism**, not cynicism — open-minded but cautious thinking.
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### **3. What Makes Something Scientific?**

A study is **scientific** when it adheres to the **scientific method**, ensuring reliability, objectivity, and empirical testing.

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### **Core Principles of Scientific Inquiry**

#### **1.     Empiricism:**

Knowledge must come from **observation and evidence**, not opinion.

## 2. **Falsifiability:**

A theory must be **testable and disprovable** (Popper, 1959).

Example: “All swans are white” is falsifiable — one black swan disproves it.

## 3. **Replication:**

Findings must be **repeatable** by other researchers.

## 4. **Objectivity:**

Personal bias must be minimized through standardized procedures.

## 5. **Systematic Observation:**

Data must be gathered in a structured, methodical way.

## 6. **Quantifiability:**

Concepts should be measurable (using operational definitions).

## 7. **Probabilistic Reasoning:**

Psychological research deals with probabilities, not certainties — human behavior varies across contexts.

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### **Example: Wundt’s First Psychological Experiment**

**Wilhelm Wundt (1832–1920)**, the “father of experimental psychology,” studied **reaction time** — how long it takes a person to respond to a stimulus.

He found that making a decision (red or green light) took longer than a simple reflex action — showing that mental processes could be **measured scientifically**.

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## **4. Steps in the Research Process**

1. **Identify a Problem / Research Question** o Example: “Does social media use cause anxiety in teenagers?”
2. **Review the Literature** o Study previous research to identify gaps and avoid repetition.
3. **Formulate a Hypothesis**
  - o A **testable statement** predicting the relationship between variables.

- Example: “Increased social media use leads to higher anxiety levels.”
  - 4. **Select Research Method** ○ Choose design: experimental, correlational, survey, case study, etc.
  - 5. **Collect Data** ○ Use valid tools: tests, interviews, surveys, observations.
  - 6. **Analyze Results** ○ Apply statistical or qualitative analysis to test hypotheses.
  - 7. **Report Findings** ○ Share results through journals, conferences, or reports.
  - 8. **Replication** ○ Others repeat the study to verify results.
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## 5. Types of Research in Psychology

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### A.

#### Experimental (Causal) Research

##### Purpose:

To establish **cause-and-effect** relationships.

##### Structure:

- **Independent Variable (IV):** Manipulated factor.
- **Dependent Variable (DV):** Measured outcome.
- **Control Group:** No treatment.
- **Experimental Group:** Receives treatment.

##### Example:

Testing whether caffeine (IV) improves memory (DV).

**Control:** Participants receive no caffeine.

**Strengths:**

- Establishes causality.
- High control and internal validity.

**Limitations:**

- Artificial settings reduce realism.
  - Not always ethical (e.g., studying trauma).
- 

**B.**

**Non-Experimental Research**

**Definition:**

No variable manipulation — researcher observes naturally occurring behavior.

**Purpose:**

Used when experiments are impossible or unethical.

**Examples:**

- Studying effects of parental divorce on children.
- Analyzing historical data (e.g., post-war mental health trends).

**Strengths:**

- Real-world context.
- Can explore phenomena over long periods.

**Limitations:**

- No control over variables.
  - Cannot infer causality.
- 

**C.****Descriptive Research****Goal:**

To describe characteristics or behaviors in detail.

**Methods:**

2. **Case Study:**
  - Deep analysis of a single individual or group.
  - Example: Freud's patient studies (e.g., "Little Hans").
3. **Naturalistic Observation:**
  - Observing behavior in its natural environment.
  - Example: Watching playground interactions.
4. **Survey Research:**
  - Using questionnaires to collect large-scale data.

**Strengths:**

- Provides detailed descriptions.

**Weakness:**

- Cannot explain causes.
-



D.

### **Correlational Research**

#### **Goal:**

To measure the **relationship between variables**.

#### **Example:**

Studying the relationship between stress and academic performance.

#### **Types of Correlation:**

- **Positive:** Both variables increase (e.g., height and weight).
- **Negative:** One increases, the other decreases (e.g., stress and sleep).
- **Zero:** No relationship.

#### **Important:**

Correlation does not imply causation.

Two variables may be related because of a third variable (confounding variable).

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E.

### **Quasi-Experimental Research**

#### **Definition:**

Similar to experiments but without random assignment.

**Example:** Comparing two existing school classes taught with different methods.

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### Used When:

Ethical or practical issues prevent random assignment.

F.

### Longitudinal vs Cross-Sectional Studies

| Type            | Description                                       | Example                                  | Advantage               | Limitation                       |
|-----------------|---------------------------------------------------|------------------------------------------|-------------------------|----------------------------------|
| Longitudinal    | Tracking memory decline in aging group over years | Studies same adults                      | Shows changes over time | Timeconsuming & expensive        |
| Cross-sectional | Compares one time                                 | Comparing teens & adults on sleep habits | Quick & inexpensive     | No cause-effect or time analysis |

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### 6. Problems and Biases in Research

- 1. Sampling Bias:**
    - Participants don't represent population.
    - Example: Only college students used → results not generalizable.
  - 2. Experimenter Bias:** ○ Researcher's expectations subtly influence outcomes.
  - 3. Demand Characteristics:** ○ Participants act how they think they should.
  - 4. Confirmation Bias:** ○ Researcher interprets data to fit their hypothesis.
  - 5. Placebo Effect:** ○ Participants improve because they believe they received treatment.
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## 6. Ethical Concerns:

- Must protect participants:
  - Informed consent
  - Confidentiality
  - Right to withdraw
  - No physical or psychological harm

## 7. Quantitative vs Qualitative Research

| Feature          | Quantitative Research               | Qualitative Research              |
|------------------|-------------------------------------|-----------------------------------|
| <b>Focus</b>     | Numbers and measurements            | Meanings and experiences          |
| <b>Goal</b>      | Test hypotheses, generalize results | Explore depth, understand context |
| <b>Approach</b>  | Objective, structured               | Subjective, open-ended            |
| <b>Data Type</b> | Numerical (scores, scales)          | Textual (words, images)           |
| <b>Analysis</b>  | Statistical                         | Thematic / narrative              |
| <b>Example</b>   | Survey on stress levels             | Interview on coping experiences   |
| <b>Strength</b>  | Reliable, generalizable             | Deep, detailed insights           |
| <b>Weakness</b>  | May ignore context                  | Hard to generalize                |

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### Quantitative Methods Include:

- Experiments
  - Surveys
  - Questionnaires
  - Database analysis
  - Statistical modeling
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**Qualitative Methods Include:**

- In-depth interviews
- Focus group discussions
- Participant observations
- Case studies
- Document analysis

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**8. Folk Psychology vs Scientific Psychology****Folk Psychology:**

Refers to everyday beliefs and “common sense” ideas people have about behavior and the mind.

These are often influenced by culture, experience, and intuition rather than science.

**Scientific Psychology:**

Uses the scientific method to test ideas with evidence and reject unsupported claims.

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**Common Myths of Folk Psychology**

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| <b>Claim</b>      | <b>Scientific Finding</b>       |
|-------------------|---------------------------------|
| Opposites attract | Similarity predicts attraction. |

|                                         |                                                               |
|-----------------------------------------|---------------------------------------------------------------|
| Familiarity breeds contempt             | Familiarity usually increases liking.                         |
| More people = more help                 | Bystander effect: people are less likely to act.              |
| Visual vs verbal learners               | Learning styles theory lacks evidence.                        |
| Hypnosis is fake                        | Has real psychological effects in pain reduction and therapy. |
| Subliminal ads work                     | No strong evidence.                                           |
| Mozart makes babies smarter             | Only short-term effects.                                      |
| Old age = unhappiness                   | Older adults often report higher life satisfaction.           |
| Trust your first answer                 | Changing answers can improve accuracy.                        |
| Stress causes ulcers                    | Caused by bacteria (H. pylori).                               |
| Positive thinking cures cancer          | No direct causal link.                                        |
| Parenting solely determines personality | Genetics and peers also shape traits.                         |
| Inkblot tests reveal personality        | Low reliability and validity.                                 |
| Interviews predict job success          | Structured tests are better predictors.                       |

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## 9. Cultural Reflection: Psychology Beyond the West

The lecture ends by reminding us that most “folk psychology” ideas come from the **Western world**.

Students are encouraged to **reflect on local cultural beliefs** — for example:

- “Left-handed people are unlucky.”
- “Dreams always predict the future.”
- “Mental illness is caused by evil spirits.”

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Psychology encourages evaluating such beliefs **scientifically**, not dismissively — combining **critical thinking** with **cultural awareness**.

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## FINAL SUMMARY TABLE (Exam Quick Revision)

| Topic                        | Core Idea                          | Key Example                            |
|------------------------------|------------------------------------|----------------------------------------|
| <b>Critical Thinking</b>     | Evaluate evidence & assumptions    | Questioning myths about mental illness |
| <b>Scientific Method</b>     | Systematic, testable, falsifiable  | Wundt's reaction time experiment       |
| <b>Experimental Research</b> | Manipulate variables to find cause | Effect of caffeine on memory           |
| <b>Non-Experimental</b>      | Observe natural situations         | Studying stress during exams           |
| <b>Descriptive</b>           | Describe behavior                  | Case study on Phineas Gage             |
| <b>Correlational</b>         | Measure relationships              | Stress vs GPA                          |
| <b>Quantitative</b>          | Numerical data, statistical        | Survey data analysis                   |
| <b>Qualitative</b>           | Narrative data, meanings           | Interview about anxiety                |
| <b>Folk Psychology</b>       | Common myths                       | "Opposites attract"                    |
| <b>Scientific Psychology</b> | Evidence-based                     | Similarity leads to liking             |

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# LECTURE 03-THE BIOLOGY OF MIND

## 1. What is the Mind?

Do you have a mind? What is it? Where is it?

- The mind is not an object you can touch.
- The mind is what your brain does.
- It is created through the brain's information processing.

## Functions of the Mind

Your mind enables:

- Thinking
- Decision-making
- Memory
- Emotions
- Imagination
- Perception (what you see, hear, feel, taste, smell)

In simple words:

Your mind is your inner world, created by the brain's activity.



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## 2. The Brain – The Central Command Center

The brain is the control system that regulates nearly everything in the body.

Major functions of the brain

- Thoughts (reasoning, planning, problem-solving)
- Memory (storing and recalling information)
- Learning (forming new connections between neurons)
- Emotions (fear, happiness, anger, love)
- Movement (coordination, balance)
- Senses (interpreting visual, auditory, taste, touch, smell signals)
- Automatic functions
- Heartbeat
- Breathing
- Digestion
- Sleep cycles
- Stress response
- Hormone regulation (e.g., puberty)

Mini-diagram (described):

Imagine a control center filled with switches — each switch manages one body function. That control center is your brain.

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### 3. The Nervous System

The nervous system is the body's communication highway.

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Two main parts:



## 1. Central Nervous System (CNS)

- Brain
- Spinal cord

## 2. Peripheral Nervous System (PNS)

- All nerves outside the brain and spinal cord
- Connects the CNS to the body (muscles, organs, skin)

### Purpose:

To send messages rapidly throughout the body using electrical and chemical signals.



## 4. Neurons – The Brain's Building Blocks

- A neuron is a nerve cell that sends and receives information.
- You have ~86 billion neurons in your brain.

### Parts of a neuron

- Dendrites – receive messages
- Cell body (soma) – processes information
- Axon – sends messages
- Synapse – tiny gap where chemical signals pass to the next neuron

### Example

When you touch a hot cup:

- 
- Sensory neurons send a pain signal → brain
  - Brain processes it

- Motor neurons send message → move your hand away

?

## 5. The Brain's Organization (3 Major Parts)

### A. The Brainstem

The oldest and most basic part of the brain.

Controls survival functions.

#### Parts of the Brainstem

1. Medulla – controls heartbeat and breathing
  2. Pons – controls sleep, facial expressions, and movement coordination
  3. Reticular formation – controls alertness and arousal
  4. Thalamus – the “sensory relay station”
- Sends sensory information (except smell) to higher brain regions

Mini-diagram (described):

A stalk-like structure at the bottom of the brain — that's the brainstem.

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### B. The Limbic System

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The emotional center of the brain.

## Key Structures

1. Amygdala – emotion (especially fear and aggression)
  - Example: Your amygdala activates when you see a snake.
2. Hippocampus – memory formation
  - Example: Helps you remember your first day of school.
3. Hypothalamus – regulates hunger, thirst, temperature, hormones
  - Commands the pituitary gland (master hormone gland)
4. Pituitary gland – releases hormones controlling growth, stress, reproduction

Mini-diagram (described):

Small structures deep inside the brain forming a ring — that is the limbic system.

?

## C. The Cerebral Cortex

The newest, most advanced part of the brain.

Responsible for complex thinking, planning, language, and consciousness.

## Functions

- Logic
  - Creativity
  - Language
  - Moral judgement
  - Planning
- 
- Problem-solving



## 6. Case Study: Phineas Gage

The most famous case in neuroscience.

### The Accident

- In 1848, a metal rod shot through Gage's cheek and out the top of his skull.
- It destroyed much of his frontal lobes.

### What happened after?

- He survived and could speak, walk, and think.
- BUT his personality changed completely:
- Became irritable
- Impulsive
- Dishonest
- Lost ability to manage emotions
- Could not plan properly

His friends said he was “no longer Gage.”

### Why is this important?

It showed that:

- The frontal lobe controls personality, decision-making, and emotion regulation.
- 
- Damage to the brain can change who we are, not just how we think.

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## 7. Language and the Brain (Aphasia)

Aphasia = language impairment from brain damage.

There are 3 important language areas:

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### 1. Broca's Area (Left Frontal Lobe)

#### Function

- Controls speaking
- Helps form words
- Coordinates speech muscles

#### Damage → Broca's Aphasia

- Person understands speech
- BUT cannot speak fluently
- Speech becomes broken or slow
- Example: "Walk... dog... store..."

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### 2. Wernicke's Area (Left Temporal Lobe)

## Function

- Understands language
- Gives meaning to words

## Damage → Wernicke's Aphasia

- Person can speak fluently
- BUT speech is meaningless
- Example sentence from a patient:

“Mother is away her working her work to get her better...”

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## 3. Angular Gyrus

### Function

- Converts written text → auditory form
- Required for reading aloud

### Damage

- Person can speak
- Person can understand speech
- BUT cannot read

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## 8. How Reading Words Works (Flow of Processing)

1 → Visual Cortex

Sees written words.

2 → Angular Gyrus

Converts written words into sound-like code.

3 → Wernicke's Area

Interprets meaning.

4 → Broca's Area

Plans speech.

5 → Motor Cortex

Moves the mouth to talk.

Example:

When reading the word “cat” aloud, your brain follows this exact pathway.

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## 9. Thinking Styles (Left vs Right Brain Tendencies)

Not scientifically rigid, but helpful to understand preferences.

Intuitive / Creative Style

- 
- Thinks in images
  - Likes stories, patterns

- Prefers hands-on learning
- Enjoys creative hobbies (art, music, writing)

#### Logical / Analytical Style

- Breaks problems into steps
- Prefers facts and sequences
- Likes instructions and guides
- Enjoys puzzles, math, strategy games

?

### 10. How to Enhance Brain Fitness (Based on Your Slides)

#### A. Sound Sleep

- Aim for 7.5 hours per night
- Sleep boosts memory, learning, mood, and focus
- Example: After studying, sleep helps store information more deeply.

?

#### B. Eat Well

Foods that support brain health:

- Nuts
- Blueberries

- 
- Whole grains
  - Avocados



- Fish oil
- Broccoli
- Tomatoes
- Seeds
- Water

Nutrients important for the brain:

- Antioxidants (protect brain cells)
- Amino acids (help with neurotransmitters)
- Vitamin E (brain protection)

?

C. Write Things Down

- Helps clear your mind
- Reduces stress
- Improves memory
- Great habit: Write tomorrow's plan the night before

?

D. Slow Down

- Focus on one task at a time
- Spend time with people
- Build friendships

- 
- Be mindful
  - Practise empathy

- Helps your brain stay calm and organized

?

#### E. Breathe

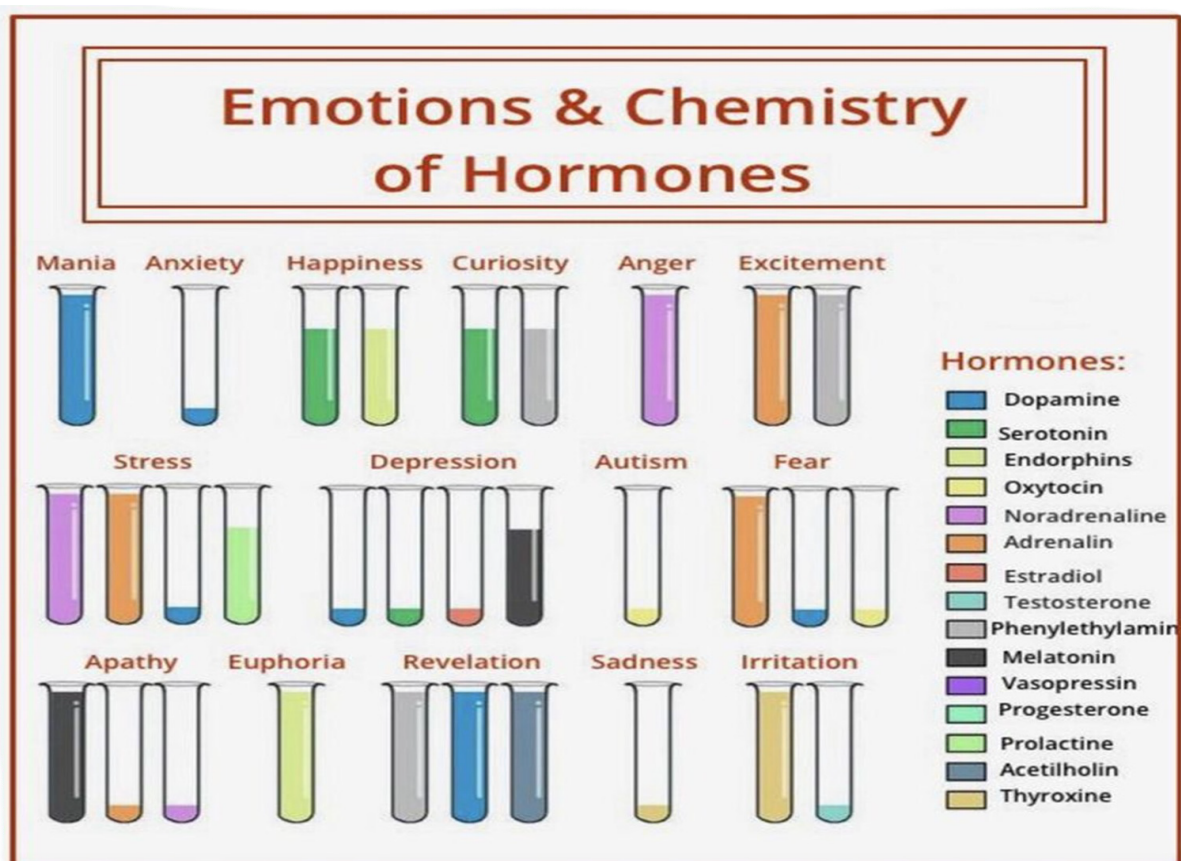
- Deep breathing reduces stress
- Improves oxygen flow to the brain
- Helps focus and emotional balance

?

#### Complete Summary (Quick Revision Section)

- Mind = what the brain does
  - Brain controls everything: thoughts → emotions → senses → memory → body functions
  - Neurons send electrical + chemical signals
  - Brainstem = survival
  - Limbic system = emotion, memory
  - Cortex = thinking, planning, language
  - Phineas Gage shows frontal lobe controls personality
  - Broca's area = speaking
  - Wernicke's area = understanding
  - Angular gyrus = reading
  - Brain fitness → sleep, nutrition, writing, slowing down, breathing
-

| Neurotransmitter    | Function                                              | Examples of Malfunctions                                                                                                                                         |
|---------------------|-------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Acetylcholine (ACh) | Enables muscle action, learning, and memory           | With Alzheimer's disease, ACh-producing neurons deteriorate.                                                                                                     |
| Dopamine            | Influences movement, learning, attention, and emotion | Excess dopamine receptor activity is linked to schizophrenia. Starved of dopamine, the brain produces the tremors and decreased mobility of Parkinson's disease. |
| Serotonin           | Affects mood, hunger, sleep, and arousal              | Undersupply is linked to depression. Prozac and some other antidepressant drugs raise serotonin levels.                                                          |
| Norepinephrine      | Helps control alertness and arousal                   | Undersupply can depress mood.                                                                                                                                    |
| Endorphins          | Lessen pain and boost mood                            | If flooded with artificial opiates, the brain may stop producing endorphins, causing intense discomfort.                                                         |



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# Lecture 04: Emotion and Motivation

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## Part 1: Emotion

---

### 1. Definition of Emotion

Emotions are **complex psychological states** that involve three distinct components:

1. **Physiological arousal** – changes in the body (heart rate, hormones, sweating, etc.)
2. **Expressive behaviors** – outward signs of emotions (facial expressions, gestures, tone of voice)
3. **Conscious experience** – the internal, subjective feeling of the emotion (how you “feel” inside)

#### In simple terms:

Emotion = a biologically-based, learned, and personally relevant reaction involving both body and mind.

#### Example:

When you see a snake:

- Your heart rate increases (physiological response)
  - You scream or jump back (expressive behavior)
  - You feel fear (subjective experience)
- 
- 

### 2. Characteristics of Emotions

- **Biological basis:** Emotions are linked to brain structures (like the amygdala and hypothalamus).
- **Personal relevance:** Triggered by situations important to the individual.
- **Automatic and immediate:** Often occur instantly, before conscious thought.
- **Shaped by learning:** Experience and culture influence how we interpret and express emotions.
- **Adaptive value:** Emotions help humans survive by guiding behavior.

---

### 3. Components of Emotion

| Component            | Description                                           | Example                                                    |
|----------------------|-------------------------------------------------------|------------------------------------------------------------|
| <b>Physiological</b> | Body's response (autonomic nervous system activation) | Heart pounding, sweating when afraid                       |
| <b>Behavioral</b>    | Expressive behavior or action                         | Smiling, frowning, crying<br>"This situation is dangerous" |
| <b>Cognitive</b>     | Thoughts, labels, and interpretation                  | → fear                                                     |

---

### 4. Early Displays of Emotion (Infants)

- Even newborns display **basic emotional expressions**, showing that emotions are **innate**.
- Basic emotions appear within the first few months:
  - Joy → smiling
  - Anger → crying or screaming
  - Disgust → wrinkled nose
  - Fear → startled reflex
  - Sadness → whimpering, tears

**Conclusion:** Emotional responses are biologically programmed rather than learned through experience.

---

### 5. Emotions Across Cultures

- **Universality:** Research by *Paul Ekman* shows that facial expressions for six basic emotions are recognized across all human cultures.
- **Cultural variations:**
  - While emotions are universal, the **rules for expressing** them (called *display rules*) differ. ◦ Example:
    - In Japan, people hide anger in public to maintain harmony.
    - In Western cultures, people express emotions more openly.

---

## 6. Six Basic Emotions (Paul Ekman)

Psychologist **Paul Ekman** identified six emotions that are **innate and universal**:

| Emotion          | Description                       | Facial Expression           | Example |
|------------------|-----------------------------------|-----------------------------|---------|
| <b>Happiness</b> | Feeling pleasure and satisfaction | Smiling, relaxed muscles    |         |
| <b>Sadness</b>   | Feeling loss or disappointment    | Drooping eyelids, tears     |         |
| <b>Fear</b>      | Perceiving threat or danger       | Wide eyes, tense body       |         |
| <b>Disgust</b>   | Feeling repulsion or aversion     | Wrinkled nose, pursed lips  |         |
| <b>Anger</b>     | Response to threat or injustice   | Frowning, clenched jaw      |         |
| <b>Surprise</b>  | Reaction to unexpected events     | Raised eyebrows, open mouth |         |

These emotions are considered the foundation for more **complex emotions** like jealousy, pride, guilt, and shame.

---

## 7. Emotion vs. Mood

| Aspect           | Emotion                                         | Mood                                                            |
|------------------|-------------------------------------------------|-----------------------------------------------------------------|
| <b>Duration</b>  | Short-term (seconds to minutes)                 | Long-term (hours to days)                                       |
| <b>Cause</b>     | Specific trigger                                | Often unclear or general                                        |
| Aspect           | Emotion                                         | Mood                                                            |
| <b>Intensity</b> | Strong and focused<br>Feeling irritable all day | Milder and diffuse<br><b>Example</b> Feeling angry after insult |

**Example:**

You might feel **anger (emotion)** after losing a game, but that may turn into a **bad mood** lasting the whole evening.

---

## 8. Functions of Emotions

Emotions are **essential for human functioning** and serve three main purposes:

### 1. Prepare us for action:

- Emotions trigger physiological responses that ready the body for response.
- Example: Fear → “fight or flight” → increases heart rate and alertness.

### 2. Shape future behavior:

- Emotional experiences help in **learning**.
- Example: Fear of failure motivates you to study harder next time.

### 3. Help in social interactions:

- Emotions communicate our feelings to others and promote bonding.
  - Example: A smile signals friendliness and approachability.
- 

## 9. Biological Basis of Emotion

- **Amygdala:** Central in fear and aggression.
  - **Hypothalamus:** Regulates physiological responses to emotion (e.g., sweating, heart rate).
  - **Prefrontal cortex:** Controls emotion regulation and decision-making.
  - **Autonomic Nervous System (ANS):** Produces changes in body state (sympathetic = arousal; parasympathetic = calm).
- 
- 
-

## Part 2: Motivation

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### 1. Definition of Motivation

**Motivation** is the **internal process** that initiates, guides, and maintains goal-directed behavior.

It is what **causes us to act** — whether to eat, study, exercise, or pursue dreams.

**Key elements of motivation:**

1. **Activation** – deciding to take action.
2. **Persistence** – continuing effort despite obstacles.
3. **Intensity** – the amount of focus and energy applied.

---

### 2. Types of Motivation

| Type                        | Description                                        | Example                                        |
|-----------------------------|----------------------------------------------------|------------------------------------------------|
| <b>Intrinsic Motivation</b> | Driven by internal satisfaction or interest.       | Studying because you love learning.            |
| <b>Extrinsic Motivation</b> | Driven by external rewards or avoiding punishment. | Studying to get good grades or avoid scolding. |

Intrinsic motivation usually leads to **long-term satisfaction** and creativity.

---

### 3. Theories of Motivation

#### A. Instinct Theory (Evolutionary Perspective)

- Proposed by **William McDougall** and influenced by **Charles Darwin**.
  - States that **behaviors are driven by instincts**, which are **inborn biological tendencies** that help survival.
- 
- Humans have instincts like curiosity, parental care, and social belonging.



**Limitation:**

- Cannot explain learned or complex behaviors like career choice or studying for exams.

**Example:**

- A newborn's rooting reflex to find milk is instinctive.
- 

**B. Drive-Reduction Theory (Clark Hull)**

- Based on the principle of **homeostasis** (balance).
- When a biological need (e.g., hunger, thirst, sleep) is unmet, it creates an **internal drive**.
- This drive **motivates** behavior to reduce tension and restore balance.

**Process:**

1. **Need:** Water
2. **Drive:** Thirst
3. **Behavior:** Drinking
4. **Homeostasis:** Restored balance

**Limitation:**

- Cannot explain behavior that increases tension (e.g., thrill-seeking, fasting).
- 

**C. Arousal Theory**

- People seek to maintain an **optimal level of arousal** — neither too low nor too high.
  - Too little arousal → boredom
- 
- Too much arousal → stress
  - Moderate arousal → peak performance (known as the *Yerkes-Dodson Law*)

### Example:

- A student performs best in exams when slightly nervous (optimal arousal).
- Too calm → lazy; too stressed → forgetful.

---

### D. Maslow's Hierarchy of Needs

- Proposed by **Abraham Maslow (1943)**.
- Suggests human needs are **arranged in a hierarchy** from basic to advanced.
- Lower needs must be met **before** higher ones become motivating.

---

### Maslow's 5 Levels of Needs

| Level | Need Type | Description | Examples |
|-------|-----------|-------------|----------|
|-------|-----------|-------------|----------|

|  |                  |                          |  |
|--|------------------|--------------------------|--|
|  | Basic biological | Air, food, water, sleep, |  |
|--|------------------|--------------------------|--|

|                      |                  |  |  |
|----------------------|------------------|--|--|
| <b>Physiological</b> | Eating, sleeping |  |  |
|----------------------|------------------|--|--|

|  |                    |  |  |
|--|--------------------|--|--|
|  | needs sex, shelter |  |  |
|--|--------------------|--|--|

|               |                           |                   |  |
|---------------|---------------------------|-------------------|--|
| <b>Safety</b> | Protection from harm, job | Safe home, stable |  |
|               | Security, stability       |                   |  |

|  |                  |        |  |
|--|------------------|--------|--|
|  | security, health | income |  |
|--|------------------|--------|--|

|                                 |                                              |                                                   |  |
|---------------------------------|----------------------------------------------|---------------------------------------------------|--|
| <b>Love &amp; Belongingness</b> | Social Friendship, intimacy, group belonging | Making friends, relationships family, being loved |  |
|---------------------------------|----------------------------------------------|---------------------------------------------------|--|

|               |                         |                                       |                         |
|---------------|-------------------------|---------------------------------------|-------------------------|
| <b>Esteem</b> | Respect and achievement | Self-respect, recognition, confidence | Getting awards, success |
|---------------|-------------------------|---------------------------------------|-------------------------|

|                           |                      |                     |  |
|---------------------------|----------------------|---------------------|--|
| <b>Self-Actualization</b> | Realizing potential, | Becoming an artist, |  |
|---------------------------|----------------------|---------------------|--|

|  |                                         |                |  |
|--|-----------------------------------------|----------------|--|
|  | Personal growth creativity, fulfillment | helping others |  |
|--|-----------------------------------------|----------------|--|

### Key idea:

---

You cannot focus on higher needs (esteem/self-actualization) if lower needs (food/safety) are unmet.

### Example:

A starving person will not care about fame until they find food.

---

## 4. Applying Maslow's Hierarchy

- Helps in **personal development, education, and workplace motivation.**
- Employers can motivate employees by ensuring:
  - o Basic salary (physiological) o Job security (safety) o Team bonding (belongingness) o Recognition (esteem) o Growth opportunities (self-actualization)

---

## 5. Comparison of Motivation Theories

| Theory          | Key Idea                                            | Motivation            | Example                         |
|-----------------|-----------------------------------------------------|-----------------------|---------------------------------|
| Source          |                                                     |                       |                                 |
| Instinct Theory | Behavior is innate<br><br>Behavior reduces internal | Biological instincts  | Babies cry when hungry          |
| Drive-Reduction | Biological drives                                   | Drinking when thirsty | tension<br><br>Doing sports for |
| Arousal Theory  | Seek optimal arousal level                          | Internal arousal      | excitement                      |
| Maslow's        | Needs must be met in                                | Hierarchy of          | Studying → Self-growth          |
|                 | <b>Hierarchy</b>                                    | order                 | needs                           |

---

## 6. Modern Understanding of Motivation

- **Cognitive factors:** Expectations, goals, and beliefs influence motivation.

- **Social factors:** Family, peers, and culture shape what we value and pursue.
  - **Emotional link:** Positive emotions (joy, excitement) increase motivation, while negative emotions (fear, anxiety) may reduce it.
- 

## 7. Connection Between Emotion and Motivation

- Emotions and motivation are **deeply linked**:
    - Emotions **trigger** motivation (fear motivates escape; love motivates care).
    - Motivation **influences** emotions (achieving goals brings joy).
  - Both involve:
    - Biological activation (nervous system) ◦ Cognitive interpretation ◦ Behavioral response
- 

## 8. Key Takeaways

- **Emotion** = Body + Mind reaction to meaningful events.
  - **Mood** = Longer emotional state.
  - **Emotions** help in adaptation, communication, and learning.
  - **Motivation** is the internal energy that drives goal-oriented behavior.
  - **Homeostasis** and **arousal levels** maintain internal balance.
  - **Maslow's hierarchy** organizes needs from basic to self-fulfillment.
  - Both emotion and motivation are **interconnected** processes that shape human behavior and personality.
- 
-

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# LECTURE 05-SENSATION & PERCEPTION

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## 1. INTRODUCTION: How We Experience the World

Your senses work **as a team** to help you navigate the environment.

Two main processes:

1. **Sensation** — receiving raw information.
  2. **Perception** — interpreting the information to make sense of the world.
- 

## 2. SENSATION

**Definition:**

Sensation is the **process by which sensory receptors and the nervous system detect and receive stimulus energy** from the environment.

**How Sensation Works (Step-by-step):**

1. A **physical stimulus** (light, sound, smell, touch, taste) hits a sensory organ.
2. Sensory receptors **absorb** this energy.
3. They **convert** the energy into **neural impulses** (electrical signals).
4. These impulses travel to the brain through neural pathways.

**Key Idea:**

---

Sensation = **Raw detection of stimulus** It does **not** yet have meaning.

---

### 3. PERCEPTION

#### Definition:

Perception is the **process of selecting, organizing, and interpreting sensory information** to form meaningful understanding of objects and events.

#### How Perception Works:

- The brain receives signals from sensation.
- It **organizes** (groups, separates, prioritizes).
- It **interprets** them based on experience, memory, and expectations.

#### Key Idea:

Perception = **Meaning-making**

Without perception, sensations are just meaningless signals.

---

### 4. HOW SENSATION & PERCEPTION WORK TOGETHER

#### Flow:

**Stimulus → Sensory Organ → Sensation → Neural Messages → Brain → Perception → Understanding**

#### Example:

- Sensation: Light hits the retina.
- Perception: Brain recognizes a face.

#### Diagram (text version):

Stimulus → Receptors → Neural Signals → Brain → Interpretation → Perception

---

## 5. THRESHOLDS

Threshold = The dividing line between detectable and undetectable stimulation.

---

### 5.1 ABSOLUTE THRESHOLD

#### Definition:

The **minimum** amount of stimulation needed to detect a stimulus **50% of the time**.

#### Examples:

- Seeing a candle flame from 30 miles away on a dark night.
  - Hearing a watch tick from 20 feet away.
  - Smelling one drop of perfume in a three-room apartment.
- 

### 5.2 DIFFERENCE THRESHOLD (Just Noticeable Difference – JND)

#### Definition:

The **smallest** change in stimulus intensity that a person can detect.

#### Important Rule — Weber's Law:

The stronger the original stimulus, the greater the change needed to notice a difference.

---

#### Examples:

- 5 → 10 lbs = noticeable

- 100 → 105 lbs = not noticeable
- TV volume 10 → 12 = noticeable
- TV volume 70 → 72 = barely noticeable

Difference threshold deals with **change**.

---

## 6. SIGNAL DETECTION THEORY (SDT)

*(Very important for exam questions)*

### Purpose:

Explains how and why people detect faint signals differently depending on psychological and physical conditions.

### Key Concepts:

- The brain always has **neural noise** (background random firing).
- A faint stimulus creates a **signal**.
- The brain must decide:

**“Is this noise or is this a real signal?”**

### Detection depends on:

1. **Stimulus intensity**
  2. **Person’s state:**
-



- o fatigue o stress
- o expectation o motivation o alertness

### Outcomes (SDT matrix):

| Actual Signal Response |                   | Example                                     |
|------------------------|-------------------|---------------------------------------------|
| Present                | Hit               | hearing a sound that is actually there      |
| Present                | Miss              | failing to hear someone call your name      |
| Absent                 | False Alarm       | thinking your phone vibrated when it didn't |
| Absent                 | Correct Rejection | recognizing that a sound was not real       |

### Application Areas:

- Decision-making
- Medical diagnosis
- Security checking
- Radar operation
- Military alert systems

---

## 7. SENSORY ADAPTATION

### Definition:

Reduced sensitivity to a constant or unchanging stimulus.

---

### Why it happens:

The brain filters out irrelevant, constant information to focus on **new and important changes**.

### Examples:

- Becoming unaware of your clothes touching your skin.
  - No longer smelling perfume after a few minutes.
  - Chef not noticing kitchen heat.
  - Worker ignoring construction noise.
- 

## 8. VISION: HOW THE BRAIN MAKES SENSE OF SIGHT

### Key Idea:

Vision is **not** just receiving light — it is the **brain's interpretation** of that light.

### Why illusions occur:

Our brain evolved shortcuts to interpret the world quickly. These shortcuts work 99% of the time, but sometimes fail → illusion.

---

## 9. THE VISUAL PROCESS (STEP-BY-STEP)

### Text Diagram:

Light → Cornea → Pupil → Lens → Retina → Optic Nerve

→ Thalamus → Visual Cortex (Occipital Lobe) **Important parts:**

- **Retina:** Converts light into neural signals.
  - **Optic Nerve:** Sends signals to brain.
  - **Visual Cortex:** Processes shapes, colors, movement, faces.
-

## 10. COLOUR VISION

### Young–Helmholtz Trichromatic Theory

#### Main Idea:

The retina has **three types of cones**:

- **Red**
- **Green**
- **Blue**

All perceived colors = combinations of these three.

#### Examples:

- Red + Green = Yellow
  - Green + Blue = Cyan • Red + Blue = Purple
- 

## 11. COLOR BLINDNESS

#### Definition:

Reduced ability to see or distinguish between certain colors.

#### Most common type:

Red–Green color blindness.

---

#### Slide Example:

People with normal vision see “74” in the dot pattern.

People with color blindness may not.

---

## 12. VISUAL INFORMATION PROCESSING

### 12.1 Feature Detection (Hubel & Wiesel)

#### Definition:

Specialized neurons in the occipital lobe respond only to specific visual features:

- Edges
- Angles
- Lines
- Movements

These neurons help identify objects quickly.

---

### 12.2 Supercell Clusters (David Perrett)

Some brain cells respond to:

- Gaze direction
- Head angle
- Posture
- Body movement

These cells help:

---

- read social cues
- understand others' attention
- detect threats

- predict movement
- 

### 13. PARALLEL PROCESSING

#### Definition:

The brain processes **multiple visual aspects at the same time**.

#### The four parts processed in parallel:

1. **Color**
2. **Motion**
3. **Form**
4. **Depth**

#### Example:

When watching a bird, you instantly know:

- its color
- shape
- how fast it's moving
- how far away it is

Your brain does all these **simultaneously**.

---

### 14. SIMPLIFIED SUMMARY OF VISUAL PROCESSING

---

#### Memory Flowchart:

Retina detects light →

Feature detectors analyze edges and motion →

Supercells integrate movement + direction →

Parallel processing handles color, form, depth, motion →

Brain combines everything into meaningful vision

---

## EXAM SURVIVAL SUMMARY (90-SECOND REVIEW)

- **Sensation** = detecting physical energy
  - **Perception** = interpreting it
  - **Absolute Threshold** = minimum detectable 50%
  - **Difference Threshold (JND)** = smallest noticeable change
  - **SDT** = detection depends on stimulus + person's state
  - **Sensory Adaptation** = lower sensitivity after constant stimulus
  - **Vision Pathway** = Retina → Brain
  - **Trichromatic Theory** = 3 cones (RGB)
  - **Color Blindness** = difficulty distinguishing colors
  - **Feature Detection** = specialized cells for edges, lines
  - **Supercells** = gaze, posture, movement
  - **Parallel Processing** = color, motion, form, depth simultaneously
- 
-

---

# LECTURE 06-Memory

*Remembering, Forgetting, and How Memory Works*

---

## 1. What Is Memory?

### Definition

Memory is the mental system that **encodes**, **stores**, and **retrieves** information. It allows us to learn from experiences, recognize familiar people or objects, solve problems, and maintain our sense of identity.

### Explanation

Memory is not like a video camera. It is an active process shaped by:

- **Bottom-up inputs** — sensory data (what we see/hear)
- **Top-down influences** — prior knowledge, expectations, culture, emotions

### Mini Diagram

Experience → Encoding → Storage → Retrieval → Behavior/Decision

### Example

You recognize your friend's voice (stored auditory memory) when they call you.

---

### Case Example

A student remembers exam content better when they connect new concepts with old knowledge — showing topdown influence.

## Memory Trick

Think **ESR** — Encoding, Storage, Retrieval (like “EdiSon’s Record”).

---

## 2. Bottom-Up vs Top-Down Influences on Memory

### Bottom-Up Processing

Information flows from **sensation** → **perception** → **memory**.

#### Example:

You hear a loud bang. You remember the sound because it was surprising.

### Top-Down Processing

Your memory is shaped by your **existing knowledge**, **culture**, and **expectations**.

#### Example:

If you believe cats are playful, you may remember a cat video as funnier than it was.

### Cultural Insight

People from collectivist cultures tend to recall **social interactions** better, while individualistic cultures recall **personal achievements** more strongly.

### Mini Diagram

BOTTOM-UP: Sensory → Meaning

TOP-DOWN: Meaning → Interpretation of Sensory

---

## 3. The Three Stages of Memory

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### 3.1 Encoding

#### Definition



The process of transforming sensory input into a meaningful memory trace.

### Types of Encoding

| Type     | Description   | Example                                 |
|----------|---------------|-----------------------------------------|
| Visual   | Images        | Remembering a logo                      |
| Auditory | Sounds        | Remembering a melody                    |
| Semantic | Meaning-based | Remembering “democracy = people’s rule” |

### Psychological Insight

Deep, meaningful processing activates more brain areas → stronger memories.

---

## 3.2 Storage

### Definition

Maintaining information over time.

### Example

Your ability to remember multiplication tables years later.

---

## 3.3 Retrieval

### Definition

Accessing stored information when needed.

### Example

You recall your ATM PIN when withdrawing money.

### Application

Self-testing strengthens retrieval pathways.

### Mini Diagram

Encoding → (Storage) → Retrieval → Performance

---

#### 4. Three-Stage Model of Memory

Sensory Memory → Short-Term Memory → Long-Term Memory

(attention)      (rehearsal)

---

#### 5. Sensory Memory

##### Definition

Brief storage of raw sensory information (visual, auditory, smell, taste, touch).

##### Types

| Type | Duration | Example |
|------|----------|---------|
|------|----------|---------|

|                        |             |                          |
|------------------------|-------------|--------------------------|
| <b>Iconic</b> (visual) | 0.3–0.5 sec | Afterimage of a sparkler |
|------------------------|-------------|--------------------------|

|                          |         |                                 |
|--------------------------|---------|---------------------------------|
| <b>Echoic</b> (auditory) | 2–4 sec | “What did you say? ... oh yes.” |
|--------------------------|---------|---------------------------------|

|                       |         |                  |
|-----------------------|---------|------------------|
| <b>Haptic</b> (touch) | 1–2 sec | Feeling of a tap |
|-----------------------|---------|------------------|

##### Key Properties

- Very accurate but fades quickly
- Helps us perceive continuity (e.g., movies appear smooth)

##### Example Activity

“What letters did you just see?” — illustrates iconic memory.

---

#### 6. Short-Term Memory (STM)

##### Definition

A temporary workspace where information is held for about **15–30 seconds**.

---

**Capacity: “7 ± 2 chunks” (Miller, 1956)**

## Chunking

Meaningful grouping of information to improve capacity.

### Example:

“17762024” → “1776 | 2024” (2 chunks instead of 8 digits)

## Digit Span Test (from lecture)

Shows increasing chunk capacity from 4 to 11 digits.

## How Prior Knowledge Helps

- Helps form meaningful chunks
- BUT may distort memory (top-down bias)

## Mini Diagram

INFO → STM (limited) → rehearsal → LTM

---

## 7. Long-Term Memory (LTM)

### Definition

Permanent, nearly limitless storage of knowledge, skills, and experiences.

---

### 7.1 Implicit (Non-declarative) Memory

#### Definition

Memory retrieved *without conscious awareness*.

#### Types & Examples

| Type | Example |
|------|---------|
|------|---------|

|            |               |
|------------|---------------|
| Procedural | Riding a bike |
|------------|---------------|

|         |                                            |
|---------|--------------------------------------------|
| Priming | Seeing “doctor” makes you think of “nurse” |
|---------|--------------------------------------------|

|                       |                           |
|-----------------------|---------------------------|
| Conditioned responses | Fear of dogs after a bite |
|-----------------------|---------------------------|

## Mini Case

A car driver automatically uses brake+clutch timing — procedural memory.

---

## 7.2 Explicit (Declarative) Memory

### Definition

Memory retrieved *with conscious effort*.

### Types

| Type | Description | Example |
|------|-------------|---------|
|------|-------------|---------|

|                 |                      |                          |
|-----------------|----------------------|--------------------------|
| <b>Episodic</b> | personal experiences | Your 10th birthday party |
|-----------------|----------------------|--------------------------|

|                 |                  |                           |
|-----------------|------------------|---------------------------|
| <b>Semantic</b> | facts, knowledge | Capital of France = Paris |
|-----------------|------------------|---------------------------|

---

### Implicit vs Explicit — Comparison Table

| Feature     | Implicit                  | Explicit      |
|-------------|---------------------------|---------------|
| Awareness   | Unconscious               | Conscious     |
| Retrieval   | Automatic                 | Effortful     |
| Examples    | Skills, priming           | Facts, events |
| Brain Areas | Basal ganglia, cerebellum | Hippocampus   |

---

## 8. Effects That Influence Memory Encoding

### 8.1 Primacy and Recency Effect

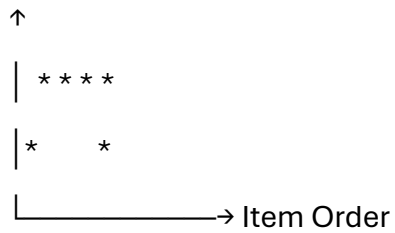
#### Explanation

In a list:

- **Primacy** → better recall of early items (LTM)
  - **Recency** → better recall of later items (STM)
- 

### Diagram

## Recall Accuracy



### Example

You remember the first and last items in a shopping list better than the middle.

---

## 8.2 Levels of Processing (Craik & Lockhart, 1972)

### Shallow Encoding

Focus on appearance (capital letters).

→ Poor memory.

### Deep Encoding

Focus on meaning.

→ Strong memory.

### Experiment Example

Participants recall more words when asked:

“Does this word relate to something alive?” than

“Is this word written in capital letters?”

### Table

| Encoding Type | Focus        | Outcome       |
|---------------|--------------|---------------|
| Shallow       | Looks, sound | Weak memory   |
| Deep          | Meaning      | Strong memory |

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## 8.3 Organization Matters

Semantically organized lists (e.g., “Career”, “Home”, “Food”) lead to better recall than random lists.

**Example:**

Grouping “LAWYER, PHYSICIAN, TEACHER” under “PROFESSIONAL” improves memory

**Mini Diagram**

Random → weak memory

Organized → strong memory

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**8.4 Context-Dependent Memory**

You recall better when in the same environment in which you learned.

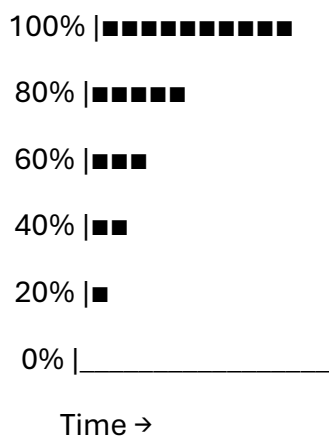
**Example:**

Studying in your room → better recall during the exam if mental context matches.

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**9. Forgetting****9.1 Ebbinghaus Forgetting Curve**

Memory loss is rapid at first, then plateaus.

**Diagram (Text-based)****Example**

You forget most of a lecture within days unless reviewed.

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**9.2 Interference Theory****Proactive Interference**

Old information blocks new information.

**Example:**

You keep typing your old password when trying to memorize a new one.

**Retroactive Interference**

New information disrupts old information.

**Example:**

Learning a new phone number makes you forget your old phone number.

**Table**

| Type | Old vs New | Example |
|------|------------|---------|
|------|------------|---------|

|           |                           |                         |
|-----------|---------------------------|-------------------------|
| Proactive | Old → interferes with new | Old password blocks new |
|-----------|---------------------------|-------------------------|

|             |                           |                             |
|-------------|---------------------------|-----------------------------|
| Retroactive | New → interferes with old | New student names erase old |
|-------------|---------------------------|-----------------------------|

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## 10. Flashbulb Memories

**Definition**

Extremely vivid, emotionally charged memories of dramatic events.

**Features**

- Caused by surprise + emotional arousal
- High confidence, but not always accurate
- Often distorted over time

**Examples**

- Remembering where you were during a major tragedy
- Recalling news of a national event

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**Cultural Insight**

Communities often share “collective flashbulb memories,” strengthening group identity.

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## 11. Improving Memory (Scientifically Proven Strategies)

### 1. Repeated, spaced study

→ Better than last-minute cramming.

### 2. Elaborative rehearsal

Connect with meaning, stories, personal examples.

### 3. Self-reference effect

Link material to your life.

#### Example:

“Proactive interference is like when I forget my new Wi-Fi password.”

### 4. Retrieval practice

Testing yourself strengthens pathways.

### 5. Reduce interference

Study in a quiet room.

### 6. Sleep

Consolidates memories into LTM.

### 7. Exercise

Boosts blood flow + neurogenesis.

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### 8. Organization

Use diagrams, tables, categories.



9. Brain-training apps

Mildly helpful for attention and working memory.

10. Healthy lifestyle

Diet, socializing, staying mentally active.

12. Final Summary Table — MEMORY (Cheat Sheet)

| Topic            | Core Idea                         | Key Example            |
|------------------|-----------------------------------|------------------------|
| Encoding         | Transforming info into memory     | Semantic > visual      |
| Storage          | Keeping info over time            | STM 15–30 sec          |
| Retrieval        | Finding stored info               | Self-testing           |
| Sensory Memory   | Brief sensory storage             | Iconic = 0.5 sec       |
| STM              | 7 ± 2 chunks                      | Phone numbers          |
| LTM              | Permanent store                   | Facts, skills          |
| Implicit         | Unconscious memory                | Riding a bike          |
| Explicit         | Conscious recall                  | Birthdays              |
| Primacy/Recency  | Serial position effects           | Remember first & last  |
| Topic            | Core Idea                         | Key Example            |
| Deep vs Shallow  | Meaning > appearance              | “Living?” > “CAPITAL?” |
| Interference     | Old ↔ New conflict                | Password confusion     |
| Flashbulb        | Emotional memories                | Tragic events          |
| Improving Memory | Spaced repetition, sleep, meaning | Study strategies       |

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# LECTURE 07: STRESS MANAGEMENT

## “Understanding Stress, Trauma, and Self-Care”

*(Full Master Note — Expanded, Clear, Human-Readable, Exam-Ready)*

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### 1. What is Stress?

#### Definition

Stress is the **psychological and physiological response** to any event or situation (a **stressor**) that disrupts the body’s equilibrium (balance).

Stress affects:

- **Mind** (thoughts)
- **Emotions** (fear, anger, worry)
- **Body** (heart rate, breathing, hormones)
- **Behavior** (avoidance, aggression, poor performance)

#### Key Characteristics

- Stress can be *real* (a danger) or *imagined* (a worry).
- The brain activates the same response for both.

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### 1.1 Why Zebras Don’t Get Ulcers (Sapolsky’s Idea)

Robert Sapolsky explains differences between animal and human stress:

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#### Animals

- Stress is **acute** (short-term)

- Triggered by **physical threats**
- Ends once danger is gone

**Example:**

A zebra runs from a lion → either survives or becomes lunch → stress ends.

---

**Humans**

- Stress is **chronic** (long-term)
- Triggered by **thoughts, fears, expectations**
- We replay events repeatedly → stress stays ON

**Example:**

Thinking about an exam happening next month can cause stress TODAY.

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**Mini Diagram**

Animal Stress → Physical → Short → Survival Based → Ends Quickly

Human Stress → Mental → Long → Future Focused → Stays Active

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**Real-Life Case Example**

Rina missed one deadline. Instead of letting it go, she keeps thinking:

“What if I fail the course? What if my parents get angry?”

→ Stress continues for weeks

→ Sleep disturbance + headaches

This is **human anticipatory stress**.

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**2. Is Stress Always Bad?**

(Weiss, Levine, Seligman)

Not all stress is harmful.

## 2.1 Types of Stress

| Type | Description | Example |
|------|-------------|---------|
|------|-------------|---------|

|                               |                      |                           |
|-------------------------------|----------------------|---------------------------|
| <b>Eustress (Good Stress)</b> | Motivates, energizes | Excitement before a match |
|-------------------------------|----------------------|---------------------------|

|                              |                    |                 |
|------------------------------|--------------------|-----------------|
| <b>Distress (Bad Stress)</b> | Overwhelms, drains | Fear of failing |
|------------------------------|--------------------|-----------------|

|                     |                   |                              |
|---------------------|-------------------|------------------------------|
| <b>Acute Stress</b> | Short and intense | Sudden braking while driving |
|---------------------|-------------------|------------------------------|

|                       |                    |                             |
|-----------------------|--------------------|-----------------------------|
| <b>Chronic Stress</b> | Long-term, harmful | Years of financial pressure |
|-----------------------|--------------------|-----------------------------|

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## 3. Psychological Modifiers of the Stress Response

These factors determine how strongly a person reacts to stress.

### 3.1 Outlets for Frustration

Healthy release mechanisms:

- Exercise
- Art / music
- Talking to someone
- Journaling
- Walking

#### Example:

A student frustrated with workload goes for a 20-minute run → stress decreases.

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### 3.2 Sense of Predictability

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Predictable events → less stress

Unpredictability → increased stress

**Example:**

Getting exam routines early reduces anxiety.

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**3.3 Sense of Control**

Feeling “I can handle this” lowers stress.

**Example:**

Making a study plan increases confidence.

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**3.4 Social Support**

Friends and family reduce emotional load.

**Case Example:**

People recovering from surgery recover faster when they have supportive family visits.

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**Mini Formula**

More Control + Predictability + Support + Outlets = Less Stress Impact

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**4. PTSD (Post-Traumatic Stress Disorder)**

PTSD occurs after experiencing or witnessing trauma.

**Definition**

A mental health condition involving intense, disturbing thoughts and emotions related to a traumatic event.

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Trauma Examples:

- Road accidents
  - Violence / abuse
  - Natural disasters
  - Sudden death of loved ones
  - War
  - Severe medical procedures
- 

## **5. Signs of PTSD (4 Symptom Clusters)**

### **A. Intrusive Symptoms**

- Flashbacks
- Nightmares
- Unwanted memories
- Emotional distress when reminded
- Physical reactions (sweating, shaking)

#### **Example:**

Hearing a loud bang and suddenly reliving an explosion scene.

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### **B. Avoidance Symptoms**

Avoiding:

- Thoughts
  - Conversations
  - Places
  - People
  - Activities connected to trauma
- 

#### **Example:**

Avoiding the road where a fatal accident happened.

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### **C. Negative Changes in Thinking & Mood**

- Negative beliefs (“I’m worthless”)
- Blaming self or others
- Persistent sadness or fear
- Loss of interest in hobbies
- Feeling emotionally numb
- Difficulty remembering parts of trauma

### **Case Example:**

A survivor of robbery stops trusting all strangers and avoids going out at night.

---

### **D. Arousal & Reactivity Symptoms**

- Easily startled
- Hypervigilance (always alert)
- Difficulty sleeping
- Irritability
- Angry outbursts
- Trouble concentrating
- Self-destructive behavior (speeding, drinking)

### **Children (under 6)**

- Re-enacting trauma during play
  - Frightening dreams
- 

## **6. Identifying Inner Resources**

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Inner resources = personal strengths that help us cope.

Examples:

- Confidence
- Patience
- Emotional intelligence
- Social skills
- Optimism
- Problem-solving
- Faith or philosophical grounding

### Example:

Positive self-talk (“I can handle this”) reduces anxiety during exams.

---

## 7. Myths About Self-Care

| Myth                          | Reality                                | Example                                               |
|-------------------------------|----------------------------------------|-------------------------------------------------------|
| Self-care is selfish          | It prevents burnout                    | Taking a short break improves productivity            |
| Self-care is temporary        | Consistent habits make lasting changes | Daily 10-min meditation                               |
| Only for women                | Needed for all humans                  | Men experience burnout too                            |
| Takes too much time           | Even 5 minutes counts                  | Stretching, deep breathing                            |
| Anything soothing is selfcare | Some soothing acts are harmful         | Drinking alcohol ≠ self-care                          |
| Same for everyone             | Personalized                           | One person likes journaling; another prefers exercise |

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## 8. Self-Care Strategies for PTSD & Stress

### 8.1 Physical Self-Care

- Exercise
- 
- Yoga
  - Breathing exercises
  - Healthy sleep patterns



**Example:**

Watering plants every morning as a calming routine.

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**8.2 Emotional Self-Care**

- Journaling
  - Reflecting on feelings
  - Forgiving self and others
  - Healthy emotional expression
- 

**8.3 Mental Self-Care**

- Reading something inspiring
  - Practicing gratitude
  - Learning new skills
- 

**8.4 Social Self-Care**

- Talking to trusted friends
  - Spending time with supportive people
  - Joining groups or communities
- 

**8.5 Boundary Setting**

Say “NO” to:

- Overcommitment
  - Energy-draining situations
  - Toxic people
- 

**Example:**

Rejecting a late-night plan before an exam.

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## 8.6 Professional Support

- Therapy
- Counseling
- Mental health workshops

### Example:

Someone with recurring flashbacks attending cognitive behavioral therapy (CBT).

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## 9. Memory Trick: “CARE PLAN” for Stress Management

C – Control (gain control)

A – Acceptance (accept what can’t be changed)

R – Release (remove emotional pressure)

E – Express (talk / journal)

P – Predictability (prepare ahead)

L – Limit stressors

A – Ask for help

N – Nourish yourself (sleep, food, rest)

---

## 10. Final Summary Table

| Topic  | Key Idea                   | Example      |
|--------|----------------------------|--------------|
| Stress | Body’s response to threats | Exam anxiety |

|                  |                                           |                                |
|------------------|-------------------------------------------|--------------------------------|
| Sapolsky         | Animals = acute; humans = chronic         | Zebra vs student               |
| PTSD             | Trauma-based disorder                     | Accident survivor              |
| Intrusion        | Flashbacks & memories                     | Nightmares after injury        |
| Avoidance        | Staying away from triggers                | Avoiding hospitals             |
| Negative Mood    | Guilt, fear, numbness                     | Disconnection from friends     |
| Hyperarousal     | Sleep issues, irritability                | Startling easily               |
| Stress Modifiers | Control, predictability, outlets, support | Study routine reducing stress  |
| Self-care        | Healthy coping                            | Meditation, reading            |
| Myths            | Misunderstanding self-care                | “Self-care is selfish” → false |

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